10329 - Spatially Resolving A HMXB Stellar Population at Cosmic Noon With Strong Lensing

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
	2	NIRCam Imaging	NIRCam Imaging	(2) Phoenix_X-ray_Arc_Image
	3	NRSIRS2-ClumpA	NIRSpec IFU Spectroscopy	(1) Phoenix_X-ray_Arc
	6	NRSIRS2-ClumpA	NIRSpec IFU Spectroscopy	(1) Phoenix_X-ray_Arc

ABSTRACT

High Mass X-ray Binaries (HMXBs) trace some of the youngest, most extreme, stellar populations. They produce large amounts of high energy emission and feedback in galaxies that are dominated by young stellar populations, a stage that describes all galaxies at some point in their evolutionary history. Furthermore, these extreme but short-lived X-ray sources can clear out the interstellar medium (mechanically), and ionize gas - both important processes for driving cosmic re-ionization. However, HMXBs and their associated stellar populations are not well-understood, especially in the distant universe. Characterizing these sources and the stellar populations that host them is crucial for a wide range of problems in observational astrophysics, from high-mass stars to galaxy evolution to re-ionization. We propose a unique study of the only highly magnified X-ray emitting dwarf starburst galaxy known. Hubble imaging and ground based spectroscopy reveal that the lensed dwarf (logM/MSun<8) galaxy hosts an X-ray bright HMXB population. The strong lensing magnification enables a zoomed-in, spatially resolved study of the HMXB-hosting region within the lensed galaxy. These data will measure the isolated emission from the HMXB population, undiluted by other regions of the galaxy, enabling a study that connects star-formation, nebular gas in the ISM, stellar populations, and HMXBs on the star cluster scale during the epoch of peak star-formation in the Universe. Joint HST UV imaging and archival MIRI imaging will multiply the science value of the JWST observations, constraining the gas and stellar populations in this lensed galaxy across 2 decades in wavelength.

OBSERVING DESCRIPTION

We request NIRCam prism IFU spectroscopy, NIRCam imaging, joint HST UV/blue imaging, and joint ground-based moderate resolution NIR spectrscopy of a strongly lensed dwarf (logM/MSun <~ 8) star forming galaxy that has a published X-ray detection. The overall goal of this program is to perform a comprehensive measurement of the spatially resolved stellar population (continuum) and ISM nebular gas (emission line) properties across the lensed, X-ray emitting galaxy, and to compare those properties to the spatially resolved X-ray emission. This analysis will reveal what stellar population and ISM properties are associated with X-ray bright (HMXB hosting) region of the galaxy.

The specific observational components of this program and their scientific purposes are outlined as follows:

1) NIRSPec IFU prism spectroscopy covering 0.6-5.3 microns (0.24-2.1 microns rest-frame) and NIRSPec IFU G140M/F100LP spectroscopy. The prism data will measure the spatially resolved stellar continuum emission from the near-UV through the NIR across the entire lensed galaxy. We will

model the complete spectrum with modern stellar population synthesis code (Prospector) to recove the star formation histories (ages), dust content, stellar masses, metallicities of each spatially resolved component of the lensed galaxy.

The G140M data will spectrally resolve numerous nebular emission lines, separating for example [Ne III] 3869 from HeI 3888 (and in combination with the prism data will enable a deblended flux measurements of the [O II] 3727,9 doublet), H-gamma from the [O III] 4363 auroral line, and H-beta from [O III] 4959A. The prism and G140M spectra, together, will be used to construct spatially resolved maps of key quantities such as the nebular reddening (using Paschen and Balmer line ratios), the instantaneous star formation rate (from Pa-alpha), and ionization parameter (ratio of [O III] 5008 to [O II] 3727), and "direct" auroral metallicity.

2) NIRCam imaging in four broad and medium bands (F090W, F210M, F360M and F410M) that are selected to avoid bright nebular emission lines, while also spanning the near-UV through near-IR. We will use these data primarily to measure the detailed morphology of the lensed galaxy (in combination with joint HST imaging, see below). We will construct GALFIT models describing all of the resolved components within the galaxy and the broad wavelength baseline available with the combination of HST and JWST will enable GALFIT models that capture all of the stellar populations (young and old) present in the lensed galaxy. The long wavelength NIRCam filers (F360M and F410M) cover emission-line free regions of continuum and will also provide valuable constraining power on the SED shape >3.5 microns (where the prism spectra SNR drops to ~3 per spectral resolution element).

3) Joint HST/UVIS imaging in F336W and F390W, which will be analyzed in combination with archival F475W imaging. These data will measure the far-UV stellar continuum emission in each spatially resolved component of the lensed galaxy. The HST photometry will be used to measure the morphology and spatial distribution of the youngest, hottest stellar populations, and will also be combined with the JWST data to model the spatially resolved stellar populations across the lensed galaxy from 0.13 to 2.1 microns (i.e., the complete spectral range dominated by stellar continua from stars of all masses).

For the background subtraction, we will use a combination of methods, including using IFU spaxels that do not contain source flux (which is a large fraction of the IFU area), direct spectra taken through the NIRSpec fixed slits, and models for the zodiacal background emission. We have spent a lot of effort using the combination of these background approaches in Cycles 1-2 data. The model method is often optimal because it injects no additional noise into the final science spectra, and the model works very well at redder wavelengths where they agree with direct (noisier) background measurements from IFU spaxels and fixed slits. The zodiacal models are not as reliable at bluer wavelengths, where we will rely on the direct background measurement from the IFU and fixed slits.

We plan for no NIRSpec LeakCal observations because subtracting them injects significant additional background noise every into the IFU cube, which prior experience has taught us is not worth the trade off for removing the occaisional spoiled region of IFU data from bright sources leaking through the MSA.

Proposal 10329 - Targets - Spatially Resolving A HMXB Stellar Population at Cosmic Noon With Strong Lensing

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	Phoenix_X-ray_Arc	RA: 23 44 46.4507 (356.1935446d) Dec: -42 43 5.00 (-42.71806d) Equinox: J2000	Proper Motion RA: 0 sec of time/yr Proper Motion Dec: 0 arcsec/yr Epoch of Position: 2000.0	
Comments: *Category=Galaxy* *Description=[Emission line galaxies, High-redshift galaxies, Starburst galaxies]* *Extended=YES*				
(2)	Phoenix_X-ray_Arc_Image	RA: 23 44 45.1162 (356.1879842d) Dec: -42 43 6.49 (-42.71847d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0 Epoch of Position: 2000.0	
Comments: *Category=Galaxy* *Description=[Emission line galaxies, Starburst galaxies]* *Extended=YES*				

Observation

Proposal 10329, Observation 2: NIRCam Imaging

Thu Jun 04 00:00:22 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	Phoenix_X-ray_Arc_Image	RA: 23 44 45.1162 (356.1879842d) Dec: -42 43 6.49 (-42.71847d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0 Epoch of Position: 2000.0	

Comments:
Category=Galaxy
Description=[Emission line galaxies, Starburst galaxies]
Extended=YES

Template

Module	Subarray	Target Placement
ALL	FULL	Detector B1 (compact source)

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEX	2	SMALL-GRID-DITHER		3

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F070W	F360M	SHALLOW4	6	1	6	6	1868.198	268954.4
2	F210M	F410M	SHALLOW4	7	1	6	6	2190.301	268954.3

Special Requirements

Aperture PA Range 4 to 67 Degrees (V3 3.65518096 to 66.65518096)
Aperture PA Range 238 to 271 Degrees (V3 237.65518096 to 270.65518096)
Aperture PA Range 276 to 296 Degrees (V3 275.65518096 to 295.65518096)
Fiducial Point Override NRCB1_FULL

Observation

Proposal 10329, Observation 3: NRSIRS2-ClumpA

Thu Jun 04 00:00:22 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	Phoenix_X-ray_Arc	RA: 23 44 46.4507 (356.1935446d)	Proper Motion RA: 0 sec of time/yr	
		Dec: -42 43 5.00 (-42.71806d)	Proper Motion Dec: 0 arcsec/yr	
		Equinox: J2000	Epoch of Position: 2000.0	

Comments:
Category=Galaxy
Description=[Emission line galaxies, High-redshift galaxies, Starburst galaxies]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	CYCLING	SMALL	1	8	

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140M/F100LP	NRSIRS2	12	2	false	true	NONE	8	16	14238.757	268954.32

Special Requirements

Aperture PA Range 196 to 208 Degrees (V3 57.02835083 to 69.02835083)

Observation

Proposal 10329, Observation 6: NRSIRS2-ClumpA

Thu Jun 04 00:00:22 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	Phoenix_X-ray_Arc	RA: 23 44 46.4507 (356.1935446d) Dec: -42 43 5.00 (-42.71806d) Equinox: J2000	Proper Motion RA: 0 sec of time/yr Proper Motion Dec: 0 arcsec/yr Epoch of Position: 2000.0	

Comments:
Category=Galaxy
Description=[Emission line galaxies, High-redshift galaxies, Starburst galaxies]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	CYCLING	SMALL	1	8	

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2	9	1	false	true	NONE	8	8	5368.712	268954.32

Special Requirements

Aperture PA Range 196 to 208 Degrees (V3 57.02835083 to 69.02835083)